



Rossmoyne Senior High School

Semester One Examination, 2023

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section Two: Calculator-assumed

If required by your examination administrator, please
place your student identification label in this box

WA student number: In figures

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In words

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Circle your Teacher's Name:

Mr Fletcher

Mrs Fraser-Jones

Mrs Murray

Mr Pisano

Mr Stephens

Time allowed for this section

Reading time before commencing work: ten minutes

Working time: one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
Total					100

Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
8	8	
9	7	
10	9	
11	8	
12	8	
13	9	
14	9	
15	7	
16	8	
17	8	
18	9	
19	8	
S2 Total	98	
S2 Wt ($\times 0.6633$)	65%	

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CALCULATOR-ASSUMED
Section Two: Calculator-assumed

3

APPLICATIONS UNIT 1
65% (98 Marks)

This section has **twelve** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

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Casual staff at a business are paid according to their age, as shown in the following table:

Age in years	16	17	18	19	20	21 or older
Hourly pay rate (\$)	14.95	15.54	20.14	22.72	25.31	27.89

The following spreadsheet is used by the business to calculate the gross weekly wage for each of their casual staff, who are paid time-and-a-half for all overtime.

Name	Age	Normal hours	Overtime hours	Gross weekly wage (\$)
Juke	18	21	3	513.57
Kiki	16	25	2	
Lacy	27	22	Q	697.25
Moya	19	P	0	522.56

(a) Calculate the gross weekly wage for Kiki. (3 marks)

(b) The business makes a contribution equal to 10.5% of each person's gross weekly wage into their superannuation fund. Determine the contribution, to the nearest cent, that Lacy will receive. (1 mark)

(c) Determine the value of P and the value of Q in the table. (4 marks)

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Information about shares in a company at the end of 2021 and 2022 is shown in the table below.

Year	Share price (\$)	Earnings per share (cents)	Annual dividend per share (cents)
2021	10.64	76	58
2022	10.45	95	68

- (a) An investor owned 8250 shares in the company. Determine the annual dividend payment, in dollars, that they received in 2021. (2 marks)

- (b) Determine the dividend yield (percentage dividend paid) in the year 2022. (2 marks)

- (c) Calculate the price-to-earnings ratio for both years and hence state in which year an investor would have paid the least for each dollar of the company's earnings. (3 marks)

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Five students have just formed a study group and want to communicate via email.

- Dave only has an email address for Eula
 - Eula only has an email address for Gugu and Heng
 - Faye only has an email address for Eula
 - Gugu only has an email address for Dave and Faye
 - Heng only has an email address for Dave
- (a) Construct matrix N to represent this communication network, with names in alphabetical order and with element $n_{i,j}$ equal to 1 if person i can send an email to person j , and equal to 0 if not. Assume that nobody can send an email to themselves. (2 marks)
- (b) Determine matrix M , where $M = N^2$. (2 marks)
- (c) State the value of element $m_{4,2}$ and explain in the context of the question what this value means. (2 marks)

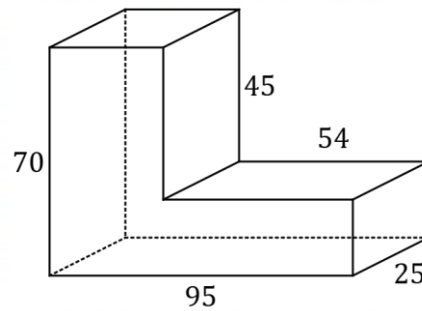
- (d) Determine $M + N$ and hence name a student who can receive emails from all the other students, either directly or via a third person. Justify your choice. (3 marks)

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APPLICATIONS UNIT 1**Question 11**

The diagram (not to scale) shows the dimensions in millimetres of a solid metal elbow used in the furniture industry.

All of the surfaces of the elbow meet at right-angles.

8**CALCULATOR-ASSUMED
(8 marks)**

- (a) Determine the total surface area of the metal elbow. (4 marks)

- (b) A new elbow is made so that it is a similar figure to the one above and the edge that was 54 mm is now 324 mm long. Determine

- (i) the length, in the new elbow, of the edge that was 45 mm long. (2 marks)

- (ii) the volume of the new elbow, given that the volume of the original elbow is 105.5cm^3 . (2 marks)

Vera is a 23-year-old student who lives away from her parent's home and is eligible for a maximum fortnightly Youth Allowance payment of \$367 when she has no other income. However, she has a casual job selling mobile phone plans from which she earns a once-off commission of 70% of the monthly cost of each plan sold.

- (a) Determine her job earnings in a week when Vera sold eight \$14.95 per month plans, four \$24.95 per month plans and two \$34.95 per month plans. (2 marks)

The Youth Allowance payments made to Vera reduce according to the following table:

Amount your payment reduces by if you earn between \$452 and \$542 per fortnight	Amount your payment reduces by if you earn more than \$542 per fortnight
50 cents for each dollar over \$452	\$45.00 plus 60 cents for each dollar over \$542

- (b) Determine the Youth Allowance payment made to Vera in a fortnight when she earned \$244.23 in the first week and \$265.37 in the second week. (3 marks)

- (c) Vera claimed that if she sold 110 of the \$14.95 per month plans in a fortnight she would receive no Youth Allowance payment. Show that her claim is correct. (3 marks)

A shipyard can buy steel ballast for a boat from three suppliers, as shown in the following table.

Supplier	Ronda Reno	Sapna Scrap	Tilla Tins
Steel ballast shape and dimensions.	Spherical with radius 8.5 cm.	Cylindrical with radius 6.6 cm and length 13 cm.	Rectangular prism of depth 12 cm, width 12 cm and length 15 cm.
Price each	\$6.50	\$5.50	\$7.50

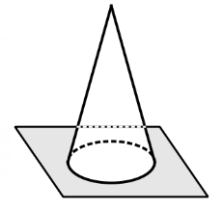
- (a) Determine the total volume of ballast the shipyard would get if they bought 20 of the steel cylinders from Sapna Scrap. (3 marks)
- (b) Determine the volume of one rectangular steel prism from Tilla Tins. (1 mark)
- (c) Determine the volume of one steel sphere from Ronda Reno. (2 marks)

- (d) Use the unit cost method to determine which supplier the shipyard should buy steel ballast from so that the cost per cubic centimetre is minimised. (3 marks)

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Diana is making miniature witch's hats out of thin card for use in a display model of a building site, and the diagram shows one of them (not to scale).

The cone has a base radius of 1.6 cm and a perpendicular height of 3 cm, and it is glued to a square base that has a side of 4.2 cm.



- (a) Calculate the circumference of the circle that glue must be applied to. (1 mark)
- (b) Calculate the slant height of the cone. (2 marks)
- (c) If Diana paints all the visible surfaces of the witch's hat, determine the total area that is painted. (3 marks)
- (d) To stop the witch's hats from being blown around, Diana fills the cone with a quick setting resin. Determine how many millilitres (mL) of resin she needs to fill 55 cones. (3 marks)

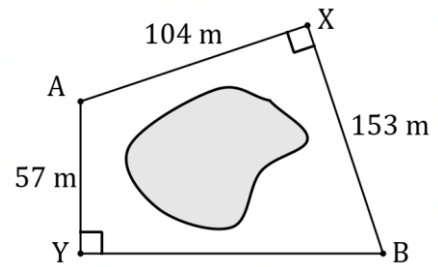
CALCULATOR-ASSUMED
Question 15

13

APPLICATIONS UNIT 1
(7 marks)

A builder must install, at a cost of \$80 per metre, an underground cable between locations A and B , as shown in the diagram (not to scale).

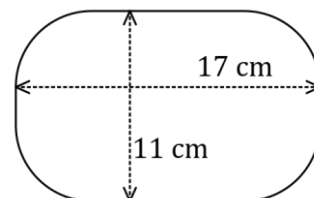
Because a lake lies between the locations, the builder must choose to lay the cable via point X or point Y , where $\angle X = \angle Y = 90^\circ$, $AX = 104$ m, $XB = 153$ m and $AY = 57$ m.



- (a) Calculate the straight-line distance between A and B . (1 mark)

- (b) Determine the least installation cost. (3 marks)

- (c) The cross-section of the cable is basically rectangular with length 17 cm, width 11 cm and rounded corners of radius 3 cm, as shown in the diagram at right (not to scale).



Determine the area of the cross-section. (3 marks)

- (a) Biba has been offered a salary increase of 5.5%. If her existing annual salary is \$75 600, determine how much this increase is worth per month. (2 marks)
- (b) After being awarded a 4% pay increase, Dove's new salary was \$67 392. Determine her salary before the increase. (2 marks)

The following is the tax table used to calculate the amount of tax to be paid for the 2021/2022 Financial year.

Taxable income	Tax on this income
0 - \$18 200	Nil
\$18 201 - \$45 000	19c for each \$1 over \$18 200
\$45 001 - \$120 000	\$5 092 plus 32.5c for each \$1 over \$45 000
\$120 001 - \$180 000	\$29 467 plus 37c for each \$1 over \$120 000
\$180 001 and over	\$51 667 plus 45c for each \$1 over \$180 000

- (c) Joy earned \$125 000 and her employer withheld \$33 000 for tax throughout the year.
- Will she get a refund, or will she have to pay extra tax?
Calculate how much that will be. (4 marks)

The price of a classic car in need of restoration was \$8800. Laine bought the car, borrowing the full purchase price from a bank at a rate of 13.2% per annum simple interest.

- (a) Determine the interest owed on this loan after six months. (2 marks)

Restoring the car cost a total of \$17 350, including parts, labour and loan interest. Laine then sold the car for \$31 600.

- (b) Calculate the profit that Laine made from the car restoration. (1 mark)

- (c) Determine the percentage increase in the value of the car from its initial value of \$8800 to its restored value of \$31 600. (2 marks)

- (d) Laine considered a loan for the purchase price of \$8800 from another bank that charged a rate of 11.5% per annum compounded annually. Determine the interest that would accumulate on this loan after two years. (3 marks)

The weight in kg of crema, espresso and mocha coffee beans sold during a week by a business in their three stores in Ellenbrook, Forrestfield and Guildford is shown in matrix A , with the Forrestfield store selling 33 kg of crema beans and the Ellenbrook store selling 10 kg of mocha beans, which was 15.5 kg less than the weight of crema beans they sold.

$$A = \begin{bmatrix} 25.5 & 13.5 & 10.0 \\ 33.0 & 22.5 & 8.5 \\ 39.5 & 19.5 & 12.5 \end{bmatrix}$$

- (a) Determine the total weight of espresso beans sold by the business during the week. (1 mark)
- (b) Describe what information the element $a_{3,1}$ represents in matrix A . (1 mark)
- (c) Determine matrix B , where $B = A \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and describe what its elements represent. (2 marks)
- (d) Write down a suitable matrix C , so that the elements of $C \times A$ represent the total weight of each type of coffee bean sold by the business during the week. (1 mark)

The business sells the crema, espresso and mocha coffee beans for \$32, \$36 and \$43 per kilogram respectively.

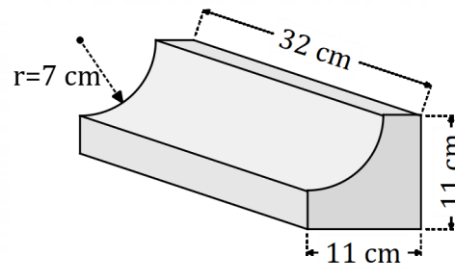
- (e) Represent the selling prices in matrix P , so that P can be multiplied with matrix A to produce a meaningful result. (1 mark)

- (f) The three matrices A , C and P can be multiplied together, not necessarily in that order, to produce a meaningful result. Clearly showing the order you use, determine the resulting matrix and describe what it represents. (3 marks)

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Question 19**(8 marks)**

A solid prism of length 32 cm has a composite cross-section as shown in the diagram (not to scale), in which a square of side 11 cm has a quarter-circle of radius 7 cm removed.



- (a) Determine the area of the composite **cross-section**. (3 marks)
- (b) Calculate the volume of the solid prism, rounding your answer to the nearest 10 cubic cm. (2 marks)
- (c) All the edges where the various surfaces meet must be trimmed with a thin silver wire. Determine the total length of wire required. (3 marks)

Supplementary page

Question number: _____

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